CASHEW: CArbon Sequestration Harnessing Energy from Waves

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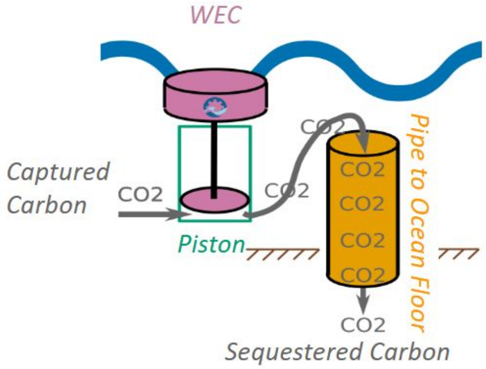
Concept:

- CASHEW uses wave energy to directly pump CO_2 into aquifers in the ocean seabed for long-term carbon sequestration
- The system couples wave energy converter motion with the pumping of CO_2 for greater efficiency (direct drive), leading to competitive a LCOC versus other resource types
- Pipe will extend from sea surface into aquifer



Approach:

- Identified two regions that met updated requirements via GIS mapping
- Developed a time-domain model of the system with wave dynamics modeled
- Modeled identified regions and region-specific parameters with the hydraulics of the WEC + pump

Next Steps:

- Implement multi-fidelity lifetime performance model
- Build and test proof of concept prototype
- Use modeling tools to aid in improving design

Blue Economy Application:	marine Carbon Dioxide Removal (mCDR)
Primary end-use/intended deployment location:	Carbon sequestration near shore globally Initial deployment focus: Gulf of America and Juan de Fuca Ridge
Estimated carbon sequestered:	0.723 Mt/yr (at \$6.30/tCO_2)
Marine energy resource type:	Wave energy